Instructions

The “mnist_dataset” zip file is given to you. Download it and extract the respective folders.

- The `.idx3-ubyte` format is a binary file format commonly used for datasets, including the MNIST dataset.
- It contains handwritten digit images from 0-9.
    - Each image is of size 28x28.
- Each file typically has a header with metadata, such as the number of images and image dimensions.
- The pixel data is flattened, meaning each image is represented as a 1D array of pixel values.
    - Each 1D array has 784 columns.

How to Load the Dataset in Jupyter Notebook:

1. Prepare the Dataset Files:

Make sure you have the `.idx3-ubyte` dataset files (e.g., train-images-idx3-ubyte, train-labels-idx1-ubyte, etc.) ready.

2. Import/ Install Necessary Libraries:

In your Jupyter Notebook, start by importing libraries like NumPy and idx2numpy for data manipulation and conversion.

Note: *you are only allowed to use the basic libraries.*

3. Load the Dataset Files:

Use the appropriate library to load the dataset files. For example, with the MNIST

dataset: import idx2numpy

train_images = idx2numpy.convert_from_file('path_to_train-images-idx3-ubyte')

train_labels = idx2numpy.convert_from_file('path_to_train-labels-idx1-ubyte')

Similarly do for test images.